

FleetManager (Monad Fleet + eLabFTW) - One-Page Summary

Who it's for (primary persona)

- Lab/research operator or engineer using eLabFTW + a fleet of devices (e.g., Raspberry Pi) for Wi-Fi sensing automation.

How it works (from repo)

Components (docker-compose.yml):

- eLabFTW web + MySQL
- monad-fleet-service: Python gRPC gateway + HTTP sidecar (/metrics, /ingest/v1/metrics)
- Agents: Raspberry Pi agent or "model-device" simulator container
- Observability: Prometheus -> Mimir -> Grafana

Data flow:

- Agent/Simulator -> gRPC :50060 -> monad-fleet-service
- monad-fleet-service -> eLabFTW REST API (https://web/api/v2) for device resources + policy from "fleet"-tagged experiments/metadata
- /metrics :9108 -> Prometheus -> remote_write -> Mimir -> Grafana
- Local state: /data/state.json (service); DATA_ROOT pending/sent (agent)

Not found in repo

- eLabFTW initial user/bootstrap credentials.

How to run (minimal)

- 1) Set docker-compose.yml: monad-fleet-service -> ELAB_API_KEY=...
- 2) docker compose up -d --build
- 3) Open:
 - eLabFTW https://localhost:8443
 - Grafana http://localhost:3000 (admin/admin)

Optional

- Prometheus http://localhost:9090, Mimir API http://localhost:9009
- End-to-end smoke test: scripts/smoke/v3_end_to_end_smoke.sh